

Srivilliputhur Municipality

AI-Based Plastic Waste & Drainage Overflow Risk Prediction System

CONSOLIDATED PROJECT DATA PACKAGE

This PDF consolidates the project information currently provided in the project notes, two source-compilation PDFs, the complete Layer 1/Layer 2 workbook, and the drainage field-survey workbook. Source classifications and limitations are preserved; synthetic data is not presented as real field data.

Included files

- Project notes: Pasted markdown.md
- Plastic / drainage source compilation: Srivilliputhur_Plastic_Waste_Drainage_Risk_Data_Sources.pdf
- GIS / rainfall / incident source compilation: Srivilliputhur_GIS_Rainfall_Incident_Data_Sources.pdf
- Complete dataset workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx
- Drainage field survey template: Srivilliputhur_Drainage_Field_Survey_Template (1)(1).xlsx

1. Project Notes / Existing Documentation

SRIVILLIPUTHUR MUNICIPALITY PROJECT — COMPLETE HANDOFF

I am continuing an academic project on:

****AI-Based Plastic Waste & Drainage Overflow Risk Prediction System for Srivilliputhur Municipality, Tamil Nadu****

I am a 3rd-year engineering student. I want to finish this project quickly, so work ****directly and step-by-step****. Do not repeatedly explain things I already know. Do not ask me to collect more data unless absolutely necessary.

I will upload the existing Excel dataset and supporting files after sending this prompt.

1. PROJECT PROBLEM

The project studies the relationship between:

****Plastic waste → rainfall → plastic entering drains → blockage → reduced drainage flow → water accumulation → overflow/waterlogging****

The system should:

- collect/organize available city drainage, rainfall, plastic/solid-waste and historical incident information;
- analyze where and when drainage problems occur;
- identify factors associated with blockage/overflow;
- build a prototype ML model to estimate LOW/MEDIUM/HIGH overflow risk;
- display risk information on a map;
- allow citizens to report drainage blockages with photos/location;
- provide information to municipality staff for inspection and maintenance;
- track maintenance and resolution;
- provide multilingual awareness information.

This is a ****prototype academic project****, not a claimed real municipal deployment.

2. VERY IMPORTANT DATA INTEGRITY RULE

Never present synthetic data as real field data.

Use these provenance categories:

VERIFIED_PUBLIC

Information supported by an identifiable official/public source.

HISTORICAL_PUBLIC

Older publicly reported information such as historical incidents/news/reports.

FIELD_DATA_REQUIRED

Information that would require actual physical field measurement but is currently unavailable.

SYNTHETIC\ DEMO

Artificial/generated data created for demonstrating the ML/application pipeline.

I cannot physically travel to Srivilliputhur for a survey, so ****do not invent or label synthetic observations as field observations****.

Do not claim that the team physically collected synthetic drain dimensions, GPS coordinates, blockage percentages, plastic accumulation levels, etc.

3. FINAL METHODOLOGY

The methodology is:

1. Collect/organize public city information.
2. Separate historical/public evidence from synthetic prototype data.
3. Validate the synthetic ML dataset.
4. Train ML only on the synthetic Layer 2 dataset.
5. Evaluate the model.
6. Analyze real/public rainfall separately.
7. Analyze historical flooding/waterlogging separately.
8. Analyze public drainage and solid-waste evidence separately.
9. Build a prototype GIS risk map.
10. Build the dashboard/application.
11. Add citizen reporting.
12. Add municipality maintenance tracking.
13. Prepare final report and PPT.

4. EXISTING WORKBOOK

The main workbook is:

****Srivilliputhur_Complete_Dataset_Layer1_Layer2.xlsx****

It should contain:

LAYER 1 — REAL/PUBLIC EVIDENCE

Sheet 1A_Historical_Incidents

Historical publicly reported incidents involving:

- waterlogging
- flooding
- sewage/drainage overflow
- heavy-rain incidents
- tank/kanmai flooding
- other relevant drainage-related events

There are approximately 12 records.

IMPORTANT:

Do NOT treat every flooding event as municipal drainage blockage.

Separate:

- municipal drain blockage/waterlogging
- natural stream/flash flooding
- tank/kanmai overflow
- Western Ghats catchment flooding

These are different phenomena.

Sheet 1B\ Verified_ Drainage

Currently contains approximately one drain-level public/secondary-source record.

Reported example:

- Ward 32
- Bharathi Nagar 3rd Street Road
- approximately 120 m
- width approximately 0.60 m
- depth approximately 0.80 m
- storm-water drain with cover slab
- one culvert
- tender/public source

Treat third-party tender information cautiously and retain the source/provenance.

Do NOT invent additional verified drains.

Sheet 1C\ Municipality_ Stats

Contains municipality-level drainage statistics and related public information.

Important official municipality figure:

****Storm-water drain length: 145.555 km****

Reported breakdown:

- Open drains: 133.40 km
- Closed drains: 12.15 km
- Approximately 79% of roads have storm-water drains.

The municipality also describes rainfall/silt influence from the western hills/mountains.

IMPORTANT:

There are older/historical reports containing figures such as 92.4 km surfaced-road drainage and 8.7 km unlined-road drainage.

Do NOT silently combine these numbers.

Record the source/year/definition for each figure.

Sheet 1D_Ward_Data

Contains 33 wards and historical/Census-era population information.

These populations are historical Census-era values and must not be described as current population unless explicitly sourced as current.

Sheet 1E_Rainfall_Portals

Contains public rainfall sources such as:

- IMD
- NWDP
- rainfall information/statistics portals

Important methodological rule:

****DO NOT merge real rainfall observations into synthetic drain rows unless there is a legitimate matching relationship based on date, location/station, and observation conditions.****

Real rainfall remains a separate public-data analysis.

Sheet 1F_Plastic_Info

Contains public/historical evidence related to:

- plastic/solid waste
- waste in drains
- littering
- drainage blockage context
- municipal/public information

Do not claim that every reported solid-waste blockage was specifically caused by plastic unless the source explicitly establishes that.

5. LAYER 2 — SYNTHETIC ML DATASET

Expected sheet:

2_Synthetic_ML_Dataset

Approximately 50 rows.

Expected columns:

```text  
Drain\_ID

Ward\_No  
Street\_Name  
Latitude  
Longitude  
Rainfall\_mm\_7day  
Rainfall\_mm\_30day  
Plastic\_Accumulation\_Level  
Plastic\_Accumulation\_Score  
Drain\_Width\_m  
Drain\_Depth\_m  
Drain\_Type  
Days\_Since\_Cleaning  
Previous\_Overflow\_Count\_1yr  
Water\_Level\_cm  
Blockage\_Percent  
Overflow\_Risk\_Label  
Data\_Source\_Type  
Notes

---

The synthetic records are for prototype ML demonstration.

They are NOT actual field measurements.

---

# 6. ML TARGET

Target:

```
```text
Overflow_Risk_Label
```

Classes:

```
```text
LOW
MEDIUM
HIGH
```

---

Candidate features:

```
```text
Rainfall_mm_7day
Rainfall_mm_30day
Plastic_Accumulation_Score
Drain_Width_m
Drain_Depth_m
Drain_Type
Days_Since_Cleaning
```

Previous_Overflow_Count_1yr
Water_Level_cm
Blockage_Percent

...

Initially exclude:

``text
Drain_ID
Street_Name
Latitude
Longitude
Data_Source_Type
Notes

...

Ward number may be retained as context or excluded from the first model depending on validation.

7. CRITICAL ML ISSUE — TARGET LEAKAGE

The synthetic target was generated using predefined risk-related variables.

Therefore, if the target was calculated directly from:

- blockage %
- plastic accumulation
- rainfall
- previous overflow

and those same variables are then given to the ML model, the model may simply learn/reconstruct the synthetic formula.

This is called **target leakage / circularity risk**.

Therefore:

1. Inspect exactly how `Overflow\_Risk\_Label` was generated.
2. Do not hide this.
3. Test for leakage.
4. If necessary, modify the prototype methodology.
5. Clearly state that model performance is synthetic-prototype performance.

DO NOT claim:

> The model predicts real Srivilliputhur drainage overflow with X% accuracy.

Instead say:

> The model demonstrates the proposed ML workflow using a synthetic prototype dataset. Its performance does not establish real-world predictive accuracy.

8. CURRENT PROJECT SEQUENCE

Follow exactly this order:

A — DATA DICTIONARY + SOURCE REFERENCE

First inspect the uploaded workbook.

Create/document:

- sheet names
- columns
- data types
- units
- meanings
- allowed values
- missing-value rules
- provenance
- source
- ML role
- limitations

Create a source reference containing:

- source name
- URL if available
- information obtained
- source status
- year/date
- limitations

Do not perform additional internet searching unless I specifically request it.

9. B — DATASET VALIDATION

After A, immediately validate the actual workbook.

Check:

Missing values

- number of missing values per column
- percentage missing

Duplicates

- duplicate rows
- duplicate IDs

Data types

- numbers stored as text
- invalid dates

- invalid categorical values

Ranges

Check for unrealistic values in:

- rainfall
- drain width
- drain depth
- water level
- blockage percentage
- cleaning days
- previous overflow count
- plastic score

Categories

Check:

```
```text
Plastic_Accumulation_Level
Drain_Type
Overflow_Risk_Label
```
```

for inconsistent spelling/capitalization.

Class balance

Count:

- LOW
- MEDIUM
- HIGH

and calculate percentages.

Label consistency

Check whether the risk label agrees with the underlying synthetic rule.

Leakage

Identify whether any feature directly or indirectly determines the target.

Dataset size

There are approximately 50 synthetic rows.

State clearly whether that is adequate for a **prototype demonstration**, while explaining that it is insufficient for claiming reliable real-world generalization.

10. C — ML MODEL

Start with:

Baseline

Logistic Regression or another simple baseline.

Main model

Random Forest Classifier.

Use an appropriate train/test strategy.

Because there are only around 50 rows:

- avoid pretending the evaluation is statistically robust;
- consider stratification if class counts permit;
- use cross-validation where appropriate;
- clearly state the limitations.

Do not select a model simply because it gives the highest score.

11. D — MODEL EVALUATION

Generate:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Feature importance

If cross-validation is used, report it clearly.

All results must be described as:

****SYNTHETIC PROTOTYPE RESULTS****

Never present them as real municipal prediction accuracy.

12. E — REAL DATA ANALYSIS

Analyze Layer 1 separately.

Rainfall

Analyze available public rainfall information.

Possible outputs:

- rainfall trends
- high rainfall periods
- seasonal patterns

Historical incidents

Analyze:

- incident frequency
- locations
- incident type
- reported causes
- temporal patterns

Drainage

Analyze:

- official drainage coverage
- open vs closed drains
- available drain-level records

Plastic/solid waste

Analyze:

- evidence of waste in drains
- reported blockage context
- public waste/plastic information

Again:

****Do not merge these into synthetic ML observations unless a valid linkage exists.****

13. F — RISK MAP

Build a prototype map showing:

- drain locations
- LOW/MEDIUM/HIGH risk
- ward boundaries if available
- citizen reports
- resolved reports

Use synthetic coordinates only as ****prototype/demo locations**** unless publicly verified.

Do not describe synthetic GPS as actual measured drain GPS.

14. G — DASHBOARD / APPLICATION

Final application should contain:

Home

- project title
- problem
- workflow
- Report a Blockage
- View Risk Map
- language selection

City Dashboard

- drains monitored
- blockage incidents
- plastic-related incidents
- overflow incidents
- open/resolved reports
- high-risk locations
- trends

GIS Risk Map

- risk levels
- drain locations
- citizen reports
- resolved reports

Drain Details

- Drain ID
- location
- drain type
- dimensions
- plastic accumulation
- previous incidents
- cleaning history
- rainfall information
- risk

ML Prediction

- input features
- predicted LOW/MEDIUM/HIGH risk
- explanation/features contributing to risk

Data Analysis

- plastic accumulation trends
- blockage trends
- rainfall vs blockage
- ward-wise incidents
- seasonal trends
- cleaning vs subsequent incidents

Citizen Reporting

- photo
- GPS/location
- issue type
- description

Report Tracking

```text

Reported

↓

Under Verification

↓

Assigned

↓

In Progress

↓

Resolved

```

Municipality Dashboard

Show:

- new reports
- high-risk drains
- photos
- locations
- historical information
- maintenance history
- pending actions
- resolved issues

Maintenance

Record:

- assigned team
- action
- cleaning date
- waste removed
- status
- before/after photos
- remarks

Awareness

Provide:

- plastic waste information
- drainage protection
- recycling/disposal information
- multilingual content

Languages:

- English
- Tamil
- Telugu
- Hindi

15. WHAT NOT TO ADD

Do NOT add technologies just to make the project sound advanced.

Do not claim:

- reinforcement learning
- IoT sensors
- live municipal integration
- automatic government decisions
- real-time prediction
- actual field measurements
- automatic crew routing
- real-time GPS tracking

unless actual data/integration exists.

Computer vision for citizen-uploaded blockage photos can be a future/optional module.

16. FINAL WORKFLOW STATEMENT

Use this as the core project workflow:

> Collect city plastic, drainage, rainfall and maintenance data → analyze blockage patterns → estimate blockage/overflow risk using ML → display risks on a city map → allow citizens to report real blockages with photos and locations → provide the information to municipalities for maintenance → track resolution → update the dataset → provide multilingual public awareness.

17. FINAL METHODOLOGY WORDING

Use this in the report:

> Public rainfall, flooding, drainage, and solid-waste information was analyzed separately as real-world evidence for Srivilliputhur Municipality. Because these observations were not linked to the synthetic drain records by date, location, and measurement conditions, they were not merged into the synthetic ML dataset. The ML prototype therefore uses clearly labeled synthetic rainfall and drain-level features. Model performance is illustrative and requires validation using real field observations.

18. IMPORTANT SOURCE/FACT CONTEXT ALREADY ESTABLISHED

Official Srivilliputhur Municipality:

<https://www.tnurbantree.gov.in/srivilliputhur/>

Official storm-water drainage page:

<https://www.tnurbantree.gov.in/srivilliputhur/storm-water-drains/>

Official ward map:

<https://www.tnurbantree.gov.in/srivilliputhur/ward-map/>

Census India:

<https://censusindia.gov.in/nada/index.php/catalog/6816>

Historical SFD report:

[https://www.susana.org/_resources/documents/default/3-4228-7-1618294487.pdf](https://www.susana.org/_resources/documents/default/3-4228-7-1618294487.pdf)

Rainfall portal:

<https://nwdp.nwic.gov.in/dataset/rainfall-daily-imd/resource/8752174f-1d17-4aaf-8058-2eb396f50157>

IMD rainfall information:

[https://mausam.imd.gov.in/responsive/rainfallinformation_swd.php](https://mausam.imd.gov.in/responsive/rainfallinformation_swd.php)

IMD rainfall statistics:

[https://mausam.imd.gov.in/responsive/rainfall_statistics.php?PAGE=1](https://mausam.imd.gov.in/responsive/rainfall_statistics.php?PAGE=1)

These should be treated according to their source type and limitations.

19. HISTORICAL INCIDENTS ALREADY COLLECTED

The existing Layer 1 dataset reportedly contains approximately 12 historical records from sources including:

- Dinamalar
- Samayam Tamil
- OneIndia Tamil
- Hindu Tamil
- Asianet
- historical SFD report

These include events involving:

- waterlogging/sewage overflow
- heavy rain
- temple flooding
- flash flooding
- tank/kanmai flooding
- road connectivity problems
- house damage

Important:

****Do not automatically classify all of these as drainage blockage incidents.****

Separate the incident mechanisms during analysis.

20. CURRENT STATUS

Data collection/search phase is considered complete for the prototype.

Do NOT send me back to search for more data unless a specific missing item prevents the next step.

The immediate task is:

FIRST:

Inspect the uploaded Excel workbook.

THEN:

Perform ****Phase A + Phase B****.

THEN:

Immediately proceed to ML.

I want to finish quickly, so be concise and action-oriented.

When you identify a problem in the dataset, tell me:

****Problem → Why it matters → Exact fix****

Do not give long theoretical explanations.

21. FILES I WILL UPLOAD

I will upload:

1. `Srivilliputhur\_Complete\_Dataset\_Layer1\_Layer2.xlsx`
2. Any supporting `paste.txt` or documentation file if needed
3. Any other project files relevant to the dataset

Use the actual uploaded files as the source of truth for the final validation.

Do not assume the workbook matches this handoff perfectly. ****Inspect it.****

After inspection, report:

****A completed / issues found****

then move directly to:

****B — Dataset Validation****

Then continue through the project without unnecessary pauses.

END OF HANDOFF

2. Plastic / drainage source compilation

Source PDF page 1

Srivilliputhur drainage-risk prototype — source compilation

Page 1

Srivilliputhur Municipality: Data Sources for an AI-Based Plastic Waste and Drainage

Blockage/Overflow Risk Prediction Prototype

Scope: publicly verifiable sources identified for Srivilliputhur Municipality, Virudhunagar District, Tamil Nadu. This report preserves uncertainty: unavailable values are explicitly marked “Not Available.” Sources in this report use the citation IDs and URLs shown in the source register.

Dataset classification

VERIFIED_PUBLIC = a currently accessible public/government/tender/news source. HISTORICAL_PUBLIC = older report/statistic useful for baseline analysis. FIELD_DATA_REQUIRED = student team must collect it. NOT_AVAILABLE = no reliable, public ward/street-level record identified.

Part

Dataset / finding

Classification

Primary source

1

Aggregate storm-water drainage network statistics; no public complete ward/street asset register.

HISTORICAL_PUBLIC

SFD Lite Report; municipal background source register

2

IMD rainfall portals and IMD daily weather reports; town-specific historic station export not verified in this search.

VERIFIED_PUBLIC

IMD rainfall information/statistics portals

3

News and sanitation-report evidence of heavy rain/flooding; ward-level incident log not found.

HISTORICAL_PUBLIC

SFD Lite Report; News18; Samayam Tamil

4

Municipal solid-waste information and town-level plastic/recyclable material notice; no drain-by-drain plastic quantity found.

VERIFIED_PUBLIC

Srivilliputhur Municipality portal

5

Town context, 33 wards, sanitation/drainage context.

HISTORICAL_PUBLIC

SFD Lite Report; Tamil Nadu Gazette

6

SFC 2023–24 SWD and culvert tender; individual BOQ measurements need tender documents.

VERIFIED_PUBLIC

Tamil Nadu eProcurement

7

Ward map, waterbody map, OpenStreetMap roads/waterways.

VERIFIED_PUBLIC

TN Urban Tree; Virudhunagar District; OSM

8

Condition, blockages, plastic accumulation, cleaning dates and local overflows require survey.

FIELD_DATA_REQUIRED

Field collection design in this report

Source PDF page 2

Srivilliputhur drainage-risk prototype — source compilation

Page 2

Part 1 — Drainage master data

A complete 33-ward drain inventory was not found in the accessible public sources. The following entries are verified only to the level stated; missing dimensions must not be inferred.

Ward

Street / location

Drain or work description

Drain type

Length

Width

Depth

Culvert information

Road /

municipal ID

Year

Source

Status

Not Av

ailable

Not Available

Municipal aggregate: storm-water

drains along surfaced roads

Storm-water

drain

92.4 km

Not

Available

Not

Available

Not Available

Not Available

Historical

municipal

statistic

Source 2 /

Source 3

HISTORICAL_

PUBLIC

Not Av

ailable

Not Available

Municipal aggregate: storm-water

drains along unlined roads

Storm-water

drain

8.7 km

Not

Available

Not

Available

Not Available

Not Available

Historical

municipal

statistic

Source 2 /
 Source 3
 HISTORICAL_
 PUBLIC
 Not Av
 ailable
 Not Available
 Roads reported without drainage
 Not Available
 12.05 km
 Not
 Available
 Not
 Available
 Not Available
 Not Available
 Historical
 municipal
 statistic
 Source 2
 HISTORICAL_
 PUBLIC
 Multipl
 e / Not
 Availa
 ble
 Srivilliputhur
 Municipality
 Construction of SWD and culvert
 under SFC 2023–24, Pack 30
 SWD + culvert
 Not
 Available
 Not
 Available
 Not
 Available
 Yes; details require
 tender BOQ
 366/2023/E1
 2023–24
 Source 6
 VERIFIED_PU
 BLIC
 5
 Ayyampatti Street
 Construction of drainage culvert
 and pave-block road
 Culvert
 Not
 Available
 Not
 Available
 Not
 Available
 Yes; dimensions

Not Available
Not Available
2023
(search
listing)
Source 7
VERIFIED_PU
BLIC
All 33 wards: coverage status
Srivilliputhur has 33 wards according to the Tamil Nadu Gazette and public local-body listings. No accessible source found a complete drainage record for Ward 1 through Ward 33. Therefore every ward's street-level drain inventory remains
FIELD_DATA_REQUIRED unless individual tender BOQs or municipal engineering registers are obtained.
Wards
Ward-level drain asset data
Public status
Action
1–33
Street, type, length, width, depth, culvert ID, construction year, condition
Not Available as a complete public dataset
Obtain municipal engineering register/RTI and validate through
field mapping.

Source PDF page 3

Srivilliputhur drainage-risk prototype — source compilation
Page 3
Part 2 — Rainfall data
IMD provides rainfall-information and rainfall-statistics portals, including district-wise daily/cumulative rainfall products. A directly downloadable, fully verified historical daily series specifically for the Srivilliputhur station was not retrieved during this
research; obtain it from IMD or a formal data request before model training.
Date / period
Location / station
Rainfall (mm)
Duration
Source
Status
Long-term baseline
Srivilliputhur
811 mm annual
average
Annual average; exact
reference period should be
checked in original report
Source 3
HISTORICAL_PUBLIC
16 Sep 2026 IMD
daily report
Srivilliputhur, Virudhunagar
district
3 cm (as reported by
IMD; equals 30 mm)
Daily reporting period; verify
report issue/date before data
use
Source 4
VERIFIED_PUBLIC
07 Jul 2021
Srivilliputhur

Not Available
 Reported as very heavy rain for
 about 1.5 hours
 Source 8
 HISTORICAL_PUBLIC
 26 Oct 2024
 Srivilliputhur and surrounding
 area
 Not Available
 Heavy rain reported
 Source 9
 HISTORICAL_PUBLIC
 17 Dec 2024
 Srivilliputhur temple area
 Not Available
 Rain/flood report
 Source 10
 HISTORICAL_PUBLIC
 Part 3 — Flooding / waterlogging incidents
 Date / year
 Location / ward / street
 Incident type
 Cause stated by source
 Rainfall stated
 Source
 Status
 2019–2021
 Multiple wards; exact wards Not
 Available
 Open-drain pollution/possible
 drainage concern
 Open drains described as carrying grey water and solid
 waste due to littering; no individual overflow event
 quantified
 Not Available
 Source 3
 HISTORICAL_PUB
 LIC
 07 Jul 2021
 Srivilliputhur town
 Heavy-rain impact; trees and
 electricity poles reportedly fell
 Not Available
 Not Available
 Source 8
 HISTORICAL_PUB
 LIC
 17 Dec 2024
 Temple area
 Rainwater surrounding temple /
 flooding report
 Not Available
 Not Available
 Source 10
 HISTORICAL_PUB
 LIC

03 Jun 2026
Near Srivilliputhur
Flash-flood report
Not Available
Not Available
Source 11
VERIFIED_PUBLIC

Source PDF page 4

Srivilliputhur drainage-risk prototype — source compilation
Page 4
Part 4 — Plastic and solid waste
No reliable public record was found that measures plastic accumulation or blockage at individual drains in Srivilliputhur. Municipal information does show that recyclable plastic and rubber were available at micro-composting/solid-waste facilities, but the notice does not provide a drain-level dataset.
Location
Ward
Waste / plastic issue
Evidence / description
Date / year
Source
Status
Srivilliputhur Municipality
Not
Available
Recyclable plastic and rubber
in municipal solid-waste system
Municipal site notice states that manure from micro-composting centres and recyclable plastic/rubber were in stock; quantity is Not Available in the search result
Current municipal
portal page
Source 12
VERIFIED_PUBLIC
Multiple locations
Not
Available
Solid waste in open drains
Sanitation report states open drains are commonly laden with household grey water and solid waste due to littering
2019–2021 report
context
Source 3
HISTORICAL_PUBLIC
All individual drains
1–33
Plastic mass, debris category,
blockage percentage
No reliable public dataset found
Not Available
Not Available
FIELD_DATA_REQUIRED
Part 5 — Municipal drainage and solid-waste context
Parameter
Value
Year / coverage

Official or primary source
Status
Municipal wards
33
2022 Gazette / current
administrative context
Source 13; Source 14
VERIFIED_PUBLIC
Municipal area
5.718 km²
Report context
Source 3
HISTORICAL_PUBLIC
Storm-water drain length on surfaced roads
92.4 km
Historical municipal statistic
Source 2
HISTORICAL_PUBLIC
Storm-water drain length on unlined roads
8.7 km
Historical municipal statistic
Source 2
HISTORICAL_PUBLIC
Road length without drainage
12.05 km
Historical municipal statistic
Source 2
HISTORICAL_PUBLIC
Drain cleaning responsibility/frequency by ward
Not Available
Not Available
No verified public operations log identified
NOT_AVAILABLE
Open-drain/closed-drain segment count
Not Available
Not Available
No verified ward/segment asset register identified
NOT_AVAILABLE

Source PDF page 5

Srivilliputhur drainage-risk prototype — source compilation

Page 5

Part 6 — Maintenance, construction and culvert works

Tender notices can establish that a proposed or procured work exists; they do not prove completion or current physical condition. Use the official tender BOQ, work order and completion certificate before treating project attributes as installed assets.

Year

Ward

Street / location

Work type

Length / quantity

Contractor / department

Tender / work ID

Source

Status

2023–24

Multiple /

Not
Available
Srivilliputhur Municipality
Construction of SWD and culvert under
SFC 2023–24 Pack 30
Not Available
MAWS / Srivilliputhur
Municipality; contractor Not
Available
366/2023/E1
Source 6
VERIFIED_PUBLIC
2023
5
Ayyampatti Street
Drainage culvert and pave-block road
Not Available
MAWS; contractor Not Available
Not Available
Source 7
VERIFIED_PUBLIC
Not Available
1–33
Not Available
Routine desilting / drain cleaning record
Not Available
Not Available
Not Available
No public log found
NOT_AVAILABLE
Part 7 — GIS and location data
Dataset / map
Coverage
Information
Source
Download / API
Status
Srivilliputhur Ward Map
Municipality; visual coverage of
wards
Ward-map image; GIS/shapefile availability Not
Available
Source 15
Viewable municipal webpage
VERIFIED_PUBLIC
Virudhunagar district waterbody map
Srivilliputhur municipality/taluk
map entry
Waterbodies and locations
Source 16
PDF map link from district site
VERIFIED_PUBLIC
OpenStreetMap
Town and surrounding region
Roads, water features and any community-mapped
drains

Source 17; Source 18
OSM API/Overpass and editable map
data; check feature completeness
VERIFIED_PUBLIC
Official municipal drain GIS layer
All wards
Drain alignment, dimensions, culverts, asset IDs
No source found
Not Available
NOT_AVAILABLE

Source PDF page 6

Srivilliputhur drainage-risk prototype — source compilation

Page 6

DATA WE MUST COLLECT OURSELVES

The following variables are central to your ML model but cannot realistically be obtained as reliable, current, ward-level public data. Record the survey date/time and observer for every observation, and retain GPS-tagged photographs as evidence.

Variable

Why unavailable publicly

Field collection method

Suggested columns

Drain ID and geometry

No complete public asset inventory with

segment dimensions

Assign a unique ID; capture GPS start/end;

tape/laser-measure width, depth and segment length

Drain_ID, Ward, Street, Start_Lat, Start_Lon, End_Lat, End_Lon,

Length_m, Width_m, Depth_m, Drain_Type, Date

Plastic accumulation

No drain-level plastic mass/volume dataset

Photograph and categorize debris; weigh safely collected

representative samples if authorised

Drain_ID, Date, Plastic_kg, Plastic_Category, Debris_Score, Photo_ID,

Observer_ID

Blockage percentage

No routine municipal blockage measurement

Estimate blocked cross-sectional area with a standard

rubric; use repeated observer checks

Drain_ID, Date, Blockage_Percent, Sediment_Depth_cm,

Water_Depth_cm, Observation_Score

Overflow event

No ward/drain event register found

Inspect after rainfall; collect timestamped photos and

citizen/worker confirmation

Event_ID, Drain_ID, Date_Time, Overflow_YN, Overflow_Depth_cm,

Photo_ID, Verification_Method

Cleaning history

No public cleaning log per drain

Request municipal logbooks and record direct

observations/interviews; mark source reliability

Drain_ID, Last_Cleaning_Date, Cleaning_Type, Source_Type, Verified_YN,

Days_Since_Cleaning

Historical local incidents

News rarely gives ward/street precision

Ward survey, municipal complaint-register request, and

structured resident interviews

Incident_ID, Ward, Street, Date, Incident_Type, Severity, Source, GPS_Lat,
GPS_Lon
Rainfall duration at site
Daily totals do not give local duration/intensity
Use IMD hourly data where available or install a
calibrated rain gauge/logger
Date_Time, Station_ID, Rainfall_mm, Duration_min, Intensity_mm_hr,
Source
ML variable availability
Proposed model input
Publicly usable now?
Recommended source / collection
Classification
Rainfall total
Partial
IMD district/station products; validate station and time resolution
VERIFIED_PUBLIC
Rainfall duration/intensity
Partial / Not Available locally
IMD hourly data if obtainable; otherwise local logger
FIELD_DATA_REQUIRED
Plastic accumulation
No
Field survey
FIELD_DATA_REQUIRED
Previous blockages
No
Field inspections / municipal complaint register
FIELD_DATA_REQUIRED
Previous overflow
Partial, non-spatial news only
Field event register; use news only as contextual evidence
FIELD_DATA_REQUIRED
Days since cleaning
No
Municipal logbook/field observations
FIELD_DATA_REQUIRED
Drain dimensions/type
Partial aggregate only
Field mapping plus BOQs/engineering register
FIELD_DATA_REQUIRED
Ward/location
Yes
Ward map, LGD/ward records, GPS survey
VERIFIED_PUBLIC
Historical waterlogging
Partial town-level reports
Create ward/street incident register through field/RTI
FIELD_DATA_REQUIRED

Source PDF page 7

Srivilliputhur drainage-risk prototype — source compilation

Page 7

Source register (exact public URLs)

Use the source register below when verifying or downloading. Where a source is a tender listing, retrieve its original tender document/BOQ from the Tamil Nadu eProcurement portal before extracting physical measurements.

ID

Source / document

Exact URL

Source 1

Srivilliputhur Municipality — Public Disclosure
<https://www.tnurbantree.tn.gov.in/srivilliputhur/public-disclosure/>

Source 2

Srivilliputhur town drainage aggregate statistic (secondary compilation; corroborate against municipal records)
<https://en.wikipedia.org/wiki/Srivilliputhur>

Source 3

SFD Lite Report: Srivilliputhur, India (KARE / SuSanA PDF)
https://www.susana.org/_resources/documents/default/3-4228-7-1618294487.pdf

Source 4

India Meteorological Department — Rainfall Information
<https://mausam.imd.gov.in/responsive/rainfallinformation.php>

Source 5

India Meteorological Department — Rainfall Statistics
https://mausam.imd.gov.in/responsive/rainfall_statistics.php?PAGE=1

Source 6

Tamil Nadu eProcurement — Construction of SWD and Culvert in Srivilliputhur Municipality, SFC 2023–24
[https://tntenders.gov.in/nicgep/app?component=\\$DirectLink&page=FrontEndViewTender&service=direct&sp=S0ogkhgbRyi+5RcnWvJLlrQ==](https://tntenders.gov.in/nicgep/app?component=$DirectLink&page=FrontEndViewTender&service=direct&sp=S0ogkhgbRyi+5RcnWvJLlrQ==)

Source 7

IndianTenders — drainage culvert and pave-block road, Ward 5 Ayyampatti Street (secondary tender listing)
<https://www.indiantenders.in/tenderdetail/construction-drainage-culvert-and-pave-block-road-ward-no-5-ayyampatti-street-srivilliputhur-municipality-c365864/>

Source 8

Samayam Tamil — heavy rain report, Srivilliputhur, 7 July 2021
<https://tamil.samayam.com/tamil-nadu/virudhunagar/very-heavy-rain-at-srivilliputhur-for-one-and-half-hour-virudhunagar-powder-cut-trees-fell-at-roads-around-20-eb-post-destroyed/articleshow/84198741.cms>

Source 9

Samayam Tamil — heavy rain report, 26 October 2024
<https://tamil.samayam.com/tamil-nadu/virudhunagar/heavy-rain-in-srivilliputhur-and-surrounding-areas/articleshow/114606791.cms>

Source 10

News18 Tamil — temple flooding/rainwater report, 17 December 2024
<https://tamil.news18.com/videos/tamil-nadu/srivilliputhur-temple-flood-rainwater-surrounds-the-temple-devotees-in-distress-rain-updates-tn-rain-1680120.html>

Source 11

News18 Tamil/Valaioosai — flash-flood video listing, 3 June 2026
<https://www.valaioosai.com/video/srivilliputhur-flash-flood-tn-rain-update-xXsj1nO0Qs>

Source 12

Srivilliputhur Municipality portal — solid-waste/recyclable plastic notice
<https://www.tnurbantree.tn.gov.in/srivilliputhur/>

Source 13

Tamil Nadu Government Gazette — ULB ward reservation document mentioning Srivilliputhur 33 wards
<https://www.tnurbantree.tn.gov.in/tiruvannamalai/wp-content/uploads/sites/11/2022/01/ULB-Election-2022-Reservation-of-WardsChairman.pdf>

Source 14

LocalBodyData — Srivilliputhur Municipality ward list (secondary

reference)
<https://localbodydata.com/municipality-srivilliputhur-252932>
Source 15
Srivilliputhur Municipality Ward Map
<https://www.tnurbantree.tn.gov.in/srivilliputhur/ward-map/>
Source 16
Virudhunagar District — Waterbody maps
<https://virudhunagar.nic.in/about-district/waterbody/>
Source 17
OpenStreetMap
<https://www.openstreetmap.org/>
Source 18
OpenStreetMap Wiki — waterway tagging
<https://wiki.openstreetmap.org/wiki/Key:waterway>

3. GIS / rainfall / incident source compilation

Source PDF page 1

Srivilliputhur GIS / rainfall / incident research baseline

Page 1

Srivilliputhur Municipality — GIS, Rainfall Time-Series and Historical Incident Data

Purpose: research baseline for an AI-Based Plastic Waste and Drainage Blockage/Overflow Risk Prediction System. This report includes only sources and values that were presented in the preceding research. It does not create synthetic rainfall, blockage, plastic, drain, or GPS data.

Important verification correction

The preceding conversational research identified useful portals but did not retrieve the complete 143-street table, a 33-ward street-to-ward crosswalk, or a Srivilliputhur station daily-rainfall CSV. Therefore, those datasets are marked Not Available / Field Data Required in this PDF rather than

reproduced as verified values. A portal's existence does not confirm its detailed records or coverage.

Classification key

VERIFIED_PUBLIC: accessible public source or portal. HISTORICAL_PUBLIC: older report/news record. FIELD_DATA_REQUIRED: collect/validate through GIS or field work. NOT_AVAILABLE: no verified granular data was retrieved.

Source PDF page 2

Srivilliputhur GIS / rainfall / incident research baseline

Page 2

1. Ward, street and GIS location data

Srivilliputhur Municipality is documented as having 33 wards. A municipal ward-map page exists, but no official downloadable boundary shapefile, complete ward-to-street table, or drain asset-ID registry was verified in this research.

Dataset / variable

Coverage

Verified information

Source

Availability / classification

Ward count

Srivilliputhur Municipality

33 wards

Source 1; Source 2

VERIFIED_PUBLIC

Ward map

Municipality

Visual ward-map page exists; geographic boundary

coordinates/shapefile were not verified

Source 3

VERIFIED_PUBLIC (map);

FIELD_DATA_REQUIRED (GIS polygons)

Ward list and LGD codes

33 wards

Public local-body listing identifies Ward 1 through Ward 33 and

LGD-code records; values should be exported directly from

original listing before use

Source 2

VERIFIED_PUBLIC

Street directory

Srivilliputhur town

OpenAlfa page provides a street guide/list based on map data;

it is not an official municipal ward crosswalk

Source 4

VERIFIED_PUBLIC

Street list/143 streets

Srivilliputhur registration area

A third-party page says it has 143 streets; full rows and ward

mapping were not retrieved and must not be treated as a

ward-wise street dataset
Source 5
NOT_AVAILABLE as ward-wise GIS dataset
Roads and mapped waterways
Town/nearby mapped area
OpenStreetMap can provide named roads, coordinates, and
any mapped waterway=drain / stream features; completeness
is community-dependent
Source 6; Source 7
VERIFIED_PUBLIC
Municipal drain IDs, road IDs,
culvert IDs
All wards
No complete public registry verified
No verified source found
NOT_AVAILABLE
Street centroid latitude/longitude
All wards
No authoritative street-coordinate table retrieved
No verified source found
FIELD_DATA_REQUIRED / derive from OSM
Recommended GIS table schema
Column
How to obtain
Status
Ward_No
Digitise official ward-map boundaries or obtain GIS file under RTI
FIELD_DATA_REQUIRED
Street_Name
OpenStreetMap / OpenAlfa; manually validate spelling in field
VERIFIED_PUBLIC + validation required
Location_Description
Municipal/tender documents and field survey
FIELD_DATA_REQUIRED
Latitude, Longitude
OSM road geometry or phone GPS observation
FIELD_DATA_REQUIRED for ground truth
Road_ID, Drain_ID, Culvert_ID
Municipal engineering register; otherwise assign project IDs after mapping
NOT_AVAILABLE publicly
Data_Source, Survey_Date, Confidence
Record provenance for every row
FIELD_DATA_REQUIRED

Source PDF page 3

Srivilliputhur GIS / rainfall / incident research baseline

Page 3

2. Detailed rainfall data

The National Water Data Portal publishes official IMD daily rainfall CSV resources at state and district level. The resource URLs below should be downloaded and filtered for Tamil Nadu/Virudhunagar; their metadata must be checked before assuming they include a station-level Srivilliputhur field.

Dataset / portal

Granularity claimed by source

Useful output for project

Exact source

Status

NWDP Rainfall DistrictWise Daily IMD
 District-wise daily rainfall
 Potential daily series for Virudhunagar district;
 inspect downloaded CSV columns and date
 coverage
 Source 8
 VERIFIED_PUBLIC
 NWDP Rainfall StateWise Daily IMD
 State-wise daily rainfall
 Tamil Nadu aggregate/state data; not proof of
 Srivilliputhur station data
 Source 9
 VERIFIED_PUBLIC
 IMD Rainfall Information
 District/state rainfall monitoring
 Current and recent daily/cumulative monitoring
 products
 Source 10
 VERIFIED_PUBLIC
 IMD Rainfall Statistics
 District/subdivision rainfall statistics
 Statistics and monitoring; station-level
 downloadable historic CSV for Srivilliputhur not
 verified here
 Source 11
 VERIFIED_PUBLIC
 IMD Chennai daily weather report
 Daily regional weather report
 May list reported station rainfall on individual
 report dates; requires extraction/validation
 Source 12
 VERIFIED_PUBLIC
 Srivilliputhur station historical daily time
 series
 Date, Rainfall_mm, Station
 Complete actual CSV/table was not retrieved in
 prior research
 Not Available
 NOT_AVAILABLE
 Verified rainfall-related observations
 Date / period
 Location
 Rainfall
 Duration / detail
 Source
 Status
 14 Sep 2026 08:30 to 15 Sep
 2026 08:30
 Srivilliputhur, Virudhunagar district
 3 cm as shown in an IMD
 regional
 daily-weather-report
 search result; retain
 original report PDF with
 dataset
 24-hour reporting window

Source 12
VERIFIED_PUBLIC
Long-term baseline
Srivilliputhur
811 mm annual average
Exact reference period must be checked in source
report before modelling

Source 13
HISTORICAL_PUBLIC
07 Jul 2021
Srivilliputhur
Not Available
Very heavy rain reported for about 1.5 hours

Source 14
HISTORICAL_PUBLIC
01 Sep 2022
Srivilliputhur
Not Available
Heavy rain reported

Source 15
HISTORICAL_PUBLIC
23 Nov 2023
Near Srivilliputhur
Not Available
Heavy rain and increased flow reported

Source 16
HISTORICAL_PUBLIC
Required actual rainfall table
Do not train the model with the event descriptions above as numeric rainfall. Download the official CSV first, retain it unchanged, then make a filtered project copy with: Date, Rainfall_mm, Geographic_Level (district/station), Station_or_District, Source_URL, Download_Date, Quality_Flag. Use district rainfall only as a district proxy; label it clearly as “Virudhunagar district”, not “Srivilliputhur station”.

Source PDF page 4

Srivilliputhur GIS / rainfall / incident research baseline
Page 4

3. Historical flooding, waterlogging and blockage records

The deeper search found rainfall/flood impacts, but not 10–30 verified municipal street-level drain-blockage records. Do not infer a drain blockage or plastic cause from a flood report unless the source explicitly says so.

Date / year

Location / ward / street

Incident

Cause explicitly documented

Rainfall

Source

Status

2019–2021 report

context

Multiple wards; wards not specified

Open drains described as carrying grey water and

solid waste

Littering stated as the reason for solid waste in open drains;

no dated overflow incident supplied

Not Available

Source 13
HISTORICAL_PUBLIC
07 Jul 2021

Srivilliputhur town
 Very heavy-rain impacts: trees and electricity
 poles reportedly fell
 Not Available
 Not Available
 Source 14
 HISTORICAL_PUBLIC
 06 Dec 2021
 Raitanpatti, Kulalar Street, Ottamadam,
 Kottaipatti, Ashok Nagar; GM Nagar mentioned
 More than 20 houses reportedly
 damaged/collapsed; report also describes water
 entering homes
 Continuous heavy rain; water and sewage mixing described
 in report. No drain-blockage cause stated.
 Not Available
 Source 17
 HISTORICAL_PUBLIC
 01 Sep 2022
 Srivilliputhur
 Heavy rain report
 Not Available
 Not Available
 Source 15
 HISTORICAL_PUBLIC
 23 Nov 2023
 Near Srivilliputhur
 Temporary road damage and increased flow
 reported
 Heavy rain; no drainage-overflow cause stated
 Not Available
 Source 16
 HISTORICAL_PUBLIC
 26 Oct 2024
 Srivilliputhur and surrounding area
 Heavy rain report
 Not Available
 Not Available
 Source 18
 HISTORICAL_PUBLIC
 02 Nov 2024
 Shenbagathoppu, Srivilliputhur Range
 Flash flood; 120 devotees reported rescued
 Heavy downpours in Western Ghats catchment according to
 forest official
 Not Available
 Source 19
 HISTORICAL_PUBLIC
 17 Dec 2024
 Srivilliputhur temple area
 Rainwater reportedly surrounded temple
 Not Available
 Not Available
 Source 20
 HISTORICAL_PUBLIC
 03 Jun 2026

Near Srivilliputhur / Shenbagathoppu context
Flash-flood video/news listing
Not Available
Not Available
Source 21
VERIFIED_PUBLIC
Incident-data limitations
None of the listed news/event records provides a verified municipal ward number, drain asset ID, drain blockage percentage, plastic quantity, or measured street-level rainfall. Shenbagathoppu/forest-range flash floods are hydrologically relevant context but should not be labelled as municipal-drain overflow incidents in the training labels.
Recommended incident register schema
Column
Meaning
Collection source
Incident_ID
Unique ID
Project-created
Date_Time
When event was observed/reported
News / municipal complaint log / field
Ward_No, Street, GPS
Precise location
Field validation / municipal register
Incident_Type
Waterlogging, overflow, blocked drain, flood, littering
Source wording; do not infer
Cause_Stated
Exact stated cause or “Not Available”
Source
Rainfall_mm, Rainfall_Source
Actual matched rainfall value and its station/district source
IMD CSV
Evidence_URL, Photo_ID,
Confidence
Traceability fields
Project records

Source PDF page 5

Srivilliputhur GIS / rainfall / incident research baseline
Page 5
4. Source register
ID
Source
Exact URL
Source 1
Tamil Nadu Gazette / ward count reference
<https://www.tnurbantree.tn.gov.in/tiruvannamalai/wp-content/uploads/sites/11/2022/01/ULB-Election-2022-Reservation-of-WardsChairman.pdf>
Source 2
LocalBodyData — Srivilliputhur Municipality wards
<https://localbodydata.com/municipality-srivilliputhur-252932>
Source 3
Srivilliputhur Municipality — Ward Map
<https://www.tnurbantree.tn.gov.in/srivilliputhur/ward-map/>
Source 4

OpenAlfa — Srivilliputhur street guide
<https://streets.openalfa.in/srivilliputhur>
Source 5

Proquiرو — Srivilliputhur street guideline-value page (third party)
<https://proquiرو.com/tools/tamil-nadu-guideline-value/srivilliputhur/srivilliputhur-street>
Source 6

OpenStreetMap
<https://www.openstreetmap.org/>
Source 7

OSM Wiki — waterway=drain definition
<https://wiki.openstreetmap.org/wiki/Tag:waterway=drain>
Source 8

NWDP — IMD district-wise daily rainfall CSV resource
<https://www.nwdp.nwic.gov.in/en/dataset/rainfall-daily-imd/resource/8752174f-1d17-4aaf-8058-2eb396f50157>
Source 9

NWDP — IMD state-wise daily rainfall CSV resource
<https://nwdp.nwic.gov.in/dataset/rainfall-daily-imd/resource/acd3023c-df96-4af8-8815-a7393cb80839>
Source 10

IMD rainfall information
https://mausam.imd.gov.in/responsive/rainfallinformation_swd.php
Source 11

IMD rainfall statistics
https://mausam.imd.gov.in/responsive/rainfall_statistics.php?PAGE=1
Source 12

IMD Chennai daily weather report
https://mausam.imd.gov.in/chennai/mcdata/daily_weather_report.pdf
Source 13

SFD Lite Report — Srivilliputhur, India
https://www.susana.org/_resources/documents/default/3-4228-7-1618294487.pdf
Source 14

Samayam Tamil — Srivilliputhur heavy rain, 7 July 2021
<https://tamil.samayam.com/tamil-nadu/virudhunagar/very-heavy-rain-at-srivilliputhur-for-one-and-half-hour-virudhunagar-power-cut-trees-fell-at-roads-around-20-eb-post-destroyed/articleshow/84198741.cms>
Source 15

Daily Thanthi — heavy rain, 1 September 2022
<https://www.dailythanthi.com/news/tamilnadu/heavy-rain-in-srivilliputhur-782882>
Source 16

Samayam Tamil — heavy rain/flow, 23 November 2023
<https://tamil.samayam.com/tamil-nadu/virudhunagar/heavy-rain-in-srivilliputhur-causing-rise-in-water-flow/articleshow/105439770.cms>
Source 17

OneIndia Tamil — heavy rain impacts, 6 December 2021
<https://tamil.oneindia.com/news/20-houses-collapsed-due-to-heavy-rains-in-srivilliputhur-441439.html>
Source 18

Samayam Tamil — heavy rain, 26 October 2024
<https://tamil.samayam.com/tamil-nadu/virudhunagar/heavy-rain-in-srivilliputhur-and-surrounding-areas/articleshow/114606791.cms>
Source 19

DT Next — flash flood rescue, 2 November 2024
<https://www.dtnext.in/news/tamilnadu/120-devotees-stuck-in-srivilliputhur-flash-floods-rescued-809729>
Source 20

News18 Tamil — temple rainwater/flooding report, 17 December 2024
<https://tamil.news18.com/videos/tamil-nadu/srivilliputhur-temple-flood-rainwater-surrounds-the-temple-devotees-in-distress-rain-updates-tn-rain-1680120.html>
Source 21

Valaioosai/News18 listing — flash flood, 3 June 2026
<https://www.valaioosai.com/video/srivilliputhur-flash-flood-tn-rain-update-xXsjl1nO0Qs>

Complete dataset workbook

Sheet: 1A_Historical_Incidents

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 12 | Columns: 11

Incident_ID	Date	Location_Street_Area	Ward_No	Incident_Type	Rainfall_mm	Reported_Cause	Impact	Source	Source_URL	Data_Source_Type
SVP-INC-001	2026-02-24	Bus Stand, Uzhavar Santhai, Periya Mariamman Kovil, Keezha Ratha Veedhi, Vadakku Ratha Veedhi junction	NOT_AVAILABLE	Waterlogging / Sewage overflow on roads	NOT_AVAILABLE	Blocked and encroached drains; silt accumulation	Sewage water flowed on roads; public inconvenience	Dinamalar	https://www.dinamalar.com/news/tamil-nadu-district-news-virudhunagar/srivilliputhur-residents-suffer-from-sewage-on-roads-due-to-rain/4158081	HISTORICAL_PUBLIC
SVP-INC-002	2024-12-17	Madavarvilagam Vaithyanatha Swamy Temple	NOT_AVAILABLE	Temple flooding / Waterlogging	NOT_AVAILABLE	Heavy rain (~2 hours)	Devotees struggled; water surrounded sanctum	Samayam Tamil	https://tamil.samayam.com/tamil-nadu/virudhunagar/srivilliputhur-and-surrounding-areas-receive-heavy-rain-in-last-night/articleshow/116419083.cms	HISTORICAL_PUBLIC
SVP-INC-003	2024-11-01	Rakakshi Amman Temple foothills stream, Western Ghats	NOT_AVAILABLE	Flash flood / Stream flooding	NOT_AVAILABLE	Heavy rain in Western Ghats catchment	150 people trapped; 3-hour rescue	OneIndia Tamil	https://tamil.oneindia.com/news/virudhunagar/fire-team-rescue-150-people-who-trapped-stream-near-srivilliputhur-in-virudhunagar-651289.html	HISTORICAL_PUBLIC
SVP-INC-004	2024-12-18	Srivilliputhur town (Mamsapuram, Vathirayiruppu, Koomapatti, Krishnan Kovil, Vanniapatti)	NOT_AVAILABLE	Heavy rain / Temple waterlogging	NOT_AVAILABLE	Continuous rain in Western Ghats	Water accumulated in temple; groundwater rose	Samayam Tamil	https://tamil.samayam.com/tamil-nadu/virudhunagar/srivilliputhur-and-surrounding-areas-receive-heavy-rain-in-last-night/articleshow/116419083.cms	HISTORICAL_PUBLIC
SVP-INC-005	2021-07-07	Srivilliputhur town (all bazaar streets)	NOT_AVAILABLE	Heavy rain / Infrastructure damage	NOT_AVAILABLE	Very heavy rain (~1.5 hours) with wind	50 trees fell; 20 EB poles destroyed; power disruption	Samayam Tamil	https://tamil.samayam.com/city/virudhunagar/very-heavy-rain-at-srivilliputhur-for-one-and-half-hour-virudhunagar-power-cut-trees-fell-at-roads-around-20-eb-post-destroyed/articleshow/84198741.cms	HISTORICAL_PUBLIC
SVP-INC-006	2021-12-06	Raitanpatti, Kulalar Street, Ottamadam, Kottaiappatti, Ashok Nagar, GM Nagar	NOT_AVAILABLE	Flooding / House collapse	NOT_AVAILABLE	Continuous heavy rain (several days)	20+ houses collapsed; 2 deaths; water entered homes; sewage mixed	OneIndia Tamil	https://tamil.oneindia.com/news/20-houses-collapse-due-to-heavy-rains-in-srivilliputhur-441439.html	HISTORICAL_PUBLIC
SVP-INC-007	2021-12-06	Srivilliputhur Vaithyanatha Swamy Temple	NOT_AVAILABLE	Temple flooding	NOT_AVAILABLE	Heavy rain (first time in 100 years)	Temple flooded; water entered premises	OneIndia Tamil	https://tamil.oneindia.com/news/20-houses-collapse-due-to-heavy-rains-in-srivilliputhur-441439.html	HISTORICAL_PUBLIC
SVP-INC-008	2021-12-06	Periyakulam Kanmai, Mottaipathan Kanmai, Ponnanganni Kanmai, Thiruvannamalai Kulam, Sengulam Kanmai, Vadamalaikurichi Kanmai	NOT_AVAILABLE	Tank flooding / Surplus release	NOT_AVAILABLE	Tanks reached full capacity	Surplus water released	OneIndia Tamil	https://tamil.oneindia.com/news/20-houses-collapse-due-to-heavy-rains-in-srivilliputhur-441439.html	HISTORICAL_PUBLIC
SVP-INC-009	2023-12-19	Sandhai Kinaru Street, Mudaliarpatti Street, Chettiyakudi Street	NOT_AVAILABLE	Flooding (tank surplus release)	183 mm in ~18 hours	Periyakulam, Vazhaikulam, Veppangulam kanmais filled; culvert opened	Water entered houses; public hardship; kanmai breach; VTO office flooded	Hindu Tamil	https://www.hindustamil.in/news/tamilnadu/1170107-in-srivilliputhur-rain-water-seeped-into-houses-due-to-an-open-culvert.html	HISTORICAL_PUBLIC
SVP-INC-010	2023-12-19	Anaiththalapatti, Achamthavirattan, Nanjarpatti, Reddiarpatti, Vilampatti	NOT_AVAILABLE	Flash flood / Road connectivity loss	183 mm in ~18 hours	Anaiththalapatti big kanmai breach	Flood water on road; 6 villages cut off; relief camps; 2 elderly rescued	Hindu Tamil	https://www.hindustamil.in/news/tamilnadu/1170107-in-srivilliputhur-rain-water-seeped-into-houses-due-to-an-open-culvert.html	HISTORICAL_PUBLIC

Incident_ID	Date	Location_Street_Area	Ward_No	Incident_Type	Rainfall_mm	Reported_Cause	Impact	Source	Source_URL	Data_Source_Type
SVP-INC-011	2022-09-26	Srivilliputhur and surrounding areas	NOT_AVAILABLE	Heavy rain	NOT_AVAILABLE	Heavy rain after heat wave	Public suffered from heat, then relief	Asianet News Tamil	https://tamil.asianetnews.com/video/tamilnadu-virut-hunagar/heavy-rain-in-srivilliputhur-and-surrounding-areas-risx2q	HISTORICAL_PUBLIC
SVP-INC-012	2019–2021 (report context)	Multiple wards (unspecified); all bazaar streets	NOT_AVAILABLE	Drain blockage / Waterlogging	NOT_AVAILABLE	Littering; drains blocked and encroached	Open drains laden with grey water and solid waste; small rain causes sewage overflow	SFD Lite Report	https://www.susana.org/_resources/documents/default/3-4228-7-1618294487.pdf	HISTORICAL_PUBLIC

Sheet: 1B_Verified_Drainage

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 1 | Columns: 11

Drain_ID	Ward_No	Street_Name	Drain_Length_m	Drain_Width_m	Drain_Depth_m	Drain_Type	Culvert_Count	Source	Source_URL	Data_Source_Type
SVP-W32-D001	32	Bharathi Nagar 3rd Street Road	120.00	0.60	0.80	Storm water drain with cover slab	1 (at chainage 73m or 171m)	MunicipalityTenders.co.in	https://www.municipalitytenders.co.in/searchresult/?pg=5&st=txt&sf=0&sd=road-construction-tamil-nadu-tenders	VERIFIED_PUBLIC

Sheet: 1C_Municipality_Stats

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 7 | Columns: 5

Parameter	Value	Year	Source	Data_Source_Type
Total storm water drain length (surfaced roads)	92.4 km	2011/2021	Wikipedia / SFD Lite Report	HISTORICAL_PUBLIC
Total storm water drain length (unlined roads)	8.7 km	2011/2021	Wikipedia / SFD Lite Report	HISTORICAL_PUBLIC
Roads with no drainage	12.05 km	2011/2021	Wikipedia	HISTORICAL_PUBLIC
Underground drainage system	Under construction (30 KLD FSTP)	2021	SFD Lite Report	HISTORICAL_PUBLIC
Percentage of roads with storm water drains	~79%	2021	TN Urban Local Bodies	VERIFIED_PUBLIC
Open drains	133.40 km	2021	TN Urban Local Bodies	VERIFIED_PUBLIC
Closed drains	12.15 km	2021	TN Urban Local Bodies	VERIFIED_PUBLIC

Sheet: 1D_Ward_Data

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 33 | Columns: 5

Ward_No	Ward_Name	Population_2011	LGD_Code	Data_Source_Type
1	Srivilliputhur Ward No - 1	2708	57099	VERIFIED_PUBLIC
2	Srivilliputhur Ward No - 2	2433	57100	VERIFIED_PUBLIC
3	Srivilliputhur Ward No - 3	2966	57101	VERIFIED_PUBLIC
4	Srivilliputhur Ward No - 4	2804	57102	VERIFIED_PUBLIC
5	Srivilliputhur Ward No - 5	2430	57103	VERIFIED_PUBLIC
6	Srivilliputhur Ward No - 6	2998	57104	VERIFIED_PUBLIC
7	Srivilliputhur Ward No - 7	2129	57105	VERIFIED_PUBLIC
8	Srivilliputhur Ward No - 8	2056	57106	VERIFIED_PUBLIC
9	Srivilliputhur Ward No - 9	5474	57107	VERIFIED_PUBLIC
10	Srivilliputhur Ward No - 10	2190	57108	VERIFIED_PUBLIC
11	Srivilliputhur Ward No - 11	2720	57109	VERIFIED_PUBLIC
12	Srivilliputhur Ward No - 12	2287	57110	VERIFIED_PUBLIC
13	Srivilliputhur Ward No - 13	1677	57111	VERIFIED_PUBLIC
14	Srivilliputhur Ward No - 14	2160	57112	VERIFIED_PUBLIC
15	Srivilliputhur Ward No - 15	2256	57113	VERIFIED_PUBLIC
16	Srivilliputhur Ward No - 16	1841	57114	VERIFIED_PUBLIC
17	Srivilliputhur Ward No - 17	1910	57115	VERIFIED_PUBLIC
18	Srivilliputhur Ward No - 18	1918	57116	VERIFIED_PUBLIC
19	Srivilliputhur Ward No - 19	2599	57117	VERIFIED_PUBLIC
20	Srivilliputhur Ward No - 20	3004	57118	VERIFIED_PUBLIC
21	Srivilliputhur Ward No - 21	2177	57119	VERIFIED_PUBLIC
22	Srivilliputhur Ward No - 22	1712	57120	VERIFIED_PUBLIC
23	Srivilliputhur Ward No - 23	1658	57121	VERIFIED_PUBLIC
24	Srivilliputhur Ward No - 24	1368	57122	VERIFIED_PUBLIC
25	Srivilliputhur Ward No - 25	1400	57123	VERIFIED_PUBLIC
26	Srivilliputhur Ward No - 26	2139	57124	VERIFIED_PUBLIC
27	Srivilliputhur Ward No - 27	2437	57125	VERIFIED_PUBLIC
28	Srivilliputhur Ward No - 28	1226	57126	VERIFIED_PUBLIC
29	Srivilliputhur Ward No - 29	2299	57127	VERIFIED_PUBLIC
30	Srivilliputhur Ward No - 30	2739	57128	VERIFIED_PUBLIC
31	Srivilliputhur Ward No - 31	1300	57129	VERIFIED_PUBLIC
32	Srivilliputhur Ward No - 32	1942	57130	VERIFIED_PUBLIC
33	Srivilliputhur Ward No - 33	2439	57131	VERIFIED_PUBLIC

Sheet: 1E_Rainfall_Portals

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 4 | Columns: 6

Portal_Name	Data_Type	Coverage	Format	Source_URL	Data_Source_Type
NWDP Rainfall DistrictWise Daily IMD	District-wise daily rainfall	All districts including Virudhunagar	CSV	https://www.nwdp.nwic.gov.in/en/dataset/rainfall-daily-imd/resource/8752174f-1d17-4aaf-8058-2eb396f50157	VERIFIED_PUBLIC
NWDP Rainfall StateWise Daily IMD	State-wise daily rainfall	Tamil Nadu stations	CSV	https://nwdp.nwic.gov.in/dataset/rainfall-daily-imd/resource/acd3023c-df96-4af8-8815-a7393cb80839	VERIFIED_PUBLIC
IMD Rainfall Information	District-wise rainfall monitoring	Interactive map, daily/cumulative	Web	https://mausam.imd.gov.in/responsive/rainfallinformation_swd.php	VERIFIED_PUBLIC
IMD Rainfall Statistics	Station-wise, district-wise, subdivision-wise	Historical + current	Web tables	https://mausam.imd.gov.in/responsive/rainfall_statistics.php?PAGE=1	VERIFIED_PUBLIC

Sheet: 1F_Plastic_Info

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 3 | Columns: 7

Location	Ward_No	Waste_Plastic_Issue	Evidence_Description	Date_Year	Source	Data_Source_Type
Multiple wards (unspecified)	NOT_AVAILABLE	Open drains blocked by solid waste (including plastic)	Most of the time, these open drains are laden with grey water from the individual households and solid waste (due to littering).	2019–2021	SFD Lite Report	HISTORICAL_PUBLIC
Ward 19, Netaji Road	19	Plastic ban enforcement; fine imposed for littering	Municipal action against plastic littering	2026	Theekkathir	VERIFIED_PUBLIC
Municipal solid-waste facility	NOT_AVAILABLE	Recyclable plastic and rubber in stock	Municipal portal notice: manure from micro-composting centres and recyclable plastic/rubber were in stock	Current	TN Urban Local Bodies	VERIFIED_PUBLIC

Sheet: 2_Synthetic_ML_Dataset

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 50 | Columns: 19

Drain_ID	Ward_No	Street_Name	Latitude	Longitude	Rainfall_mm_7day	Rainfall_mm_30day	Plastic_Accumulation_Level	Plastic_Accumulation_Score	Drain_Width_m	Drain_Depth_m	Drain_Type	Days_Since_Cleaning	Previous_Overflow_Count_1yr	Water_Level_cm	Blockage_Percent	Overflow_Risk_Label	Data_Source_Type	Notes
SVP-SYN-D001	8	Temple Street	9.516539	77.624122	55.4	208.6	MEDIUM	2	0.84	0.95	OPEN	12	1	13.1	53.2	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D002	13	Sivan Street	9.514247	77.631624	53.6	284.9	HIGH	3	0.41	1.08	OPEN	44	2	10.4	81.6	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D003	22	Kumar Nagar	9.500319	77.620418	72.4	90.9	HIGH	3	0.98	0.92	COVERED	49	0	24.3	71.4	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D004	24	Anna Street	9.502807	77.619739	13.1	63.1	HIGH	3	1.08	1.12	OPEN	59	5	34.2	18.0	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D005	23	Patel Nagar	9.514754	77.635546	39.5	151.9	MEDIUM	2	0.83	0.75	OPEN	35	5	29.1	22.6	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D006	21	VOC Street	9.503726	77.618803	34.0	237.9	NONE	0	1.15	1.13	OPEN	84	3	18.8	78.2	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D007	30	Shiva Street	9.504622	77.624166	19.7	156.5	MEDIUM	2	1.12	0.84	OPEN	18	4	22.3	65.5	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D008	8	Temple Street	9.513686	77.637802	33.8	193.6	LOW	1	1.2	0.92	COVERED	2	5	30.2	59.5	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.

Drain_ID	Ward_No	Street_Name	Latitude	Longitude	Rainfall_mm_7day	Rainfall_mm_30day	Plastic_Accumulation_Level	Plastic_Accumulation_Score	Drain_Width_m	Drain_Depth_m	Drain_Type	Days_Since_Cleaning	Previous_Overflow_Count_1yr	Water_Level_cm	Blockage_Percent	Overflow_Risk_Label	Data_Source_Type	Notes
SVP-SYN-D009	18	Chakrapani Street	9.506504	77.625338	46.5	185.4	MEDIUM	2	0.98	1.18	PARTIALLY_COVERED	65	0	35.5	28.9	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D010	33	Amman Street	9.502973	77.627348	90.7	348.4	HIGH	3	1.13	0.96	OPEN	15	2	35.8	71.5	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D011	20	Rajaji Nagar	9.499448	77.63995	59.0	168.9	LOW	1	0.46	1.06	PARTIALLY_COVERED	17	5	21.6	49.0	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D012	17	Kamalakkal Street	9.519811	77.628578	55.4	276.7	MEDIUM	2	0.56	0.79	COVERED	84	2	20.3	46.4	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D013	8	Temple Street	9.503617	77.626452	23.5	149.5	NONE	0	0.46	0.98	OPEN	5	2	7.5	24.0	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D014	32	Bharathi Nagar 3rd Street Road	9.511481	77.636084	78.9	109.4	NONE	0	0.78	0.84	OPEN	85	3	17.4	37.9	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D015	4	Byepass Road	9.499515	77.636205	50.3	143.4	NONE	0	0.83	0.68	OPEN	60	1	35.6	11.0	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D016	7	Railway Station Road	9.514304	77.631515	60.4	183.3	MEDIUM	2	0.59	0.84	OPEN	52	0	10.8	5.2	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D017	25	Rajaji Street	9.521163	77.637628	33.1	300.5	MEDIUM	2	1.03	1.0	OPEN	25	2	12.6	9.7	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.

Drain_ID	Ward_No	Street_Name	Latitude	Longitude	Rainfall_mm_7day	Rainfall_mm_30day	Plastic_Accumulation_Level	Plastic_Accumulation_Score	Drain_Width_m	Drain_Depth_m	Drain_Type	Days_Since_Cleaning	Previous_Overflow_Count_1yr	Water_Level_cm	Blockage_Percent	Overflow_Risk_Label	Data_Source_Type	Notes
SVP-SYN-D018	4	Sattur Road	9.499429	77.632604	0.0	177.2	NONE	0	1.17	0.65	OPEN	77	0	28.6	23.8	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D019	8	Main Road	9.504155	77.632863	18.3	123.1	MEDIUM	2	0.82	1.16	OPEN	41	1	14.3	15.5	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D020	20	Subramaniya Street	9.505904	77.636797	79.8	219.0	HIGH	3	1.2	0.64	OPEN	34	1	37.7	75.5	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D021	16	Bharathi Nagar	9.505125	77.628955	16.4	161.8	HIGH	3	1.05	0.92	OPEN	71	2	37.6	13.3	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D022	9	Netaji Road	9.500886	77.620676	18.6	159.1	MEDIUM	2	0.97	0.72	OPEN	34	4	22.1	77.4	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D023	4	Byepass Road	9.513857	77.638734	75.3	199.7	NONE	0	1.18	0.7	OPEN	55	4	5.3	11.0	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D024	10	Temple Road	9.498901	77.62723	9.9	152.5	NONE	0	0.69	1.16	COVERED	6	2	12.4	25.0	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D025	7	Kamarajar Street	9.517503	77.640103	4.5	241.8	NONE	0	1.14	1.12	COVERED	23	3	5.9	63.9	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D026	22	Vivekananda Street	9.518056	77.639602	55.3	121.5	MEDIUM	2	0.49	1.12	PARTIALLY_COVERED	29	1	33.6	41.8	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.

Drain_ID	Ward_No	Street_Name	Latitude	Longitude	Rainfall_mm_7day	Rainfall_mm_30day	Plastic_Accumulation_Level	Plastic_Accumulation_Score	Drain_Width_m	Drain_Depth_m	Drain_Type	Days_Since_Cleaning	Previous_Overflow_Count_1yr	Water_Level_cm	Blockage_Percent	Overflow_Risk_Label	Data_Source_Type	Notes
SVP-SYN-D027	20	Rajaji Nagar	9.503573	77.634501	22.3	218.5	NONE	0	1.02	0.81	OPEN	87	4	16.6	7.2	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D028	17	Kannadasan Street	9.512515	77.642589	90.8	205.7	LOW	1	1.03	0.86	COVERED	15	3	36.5	20.2	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D029	3	Rajaji Road	9.498042	77.641139	0.0	149.5	MEDIUM	2	0.94	0.82	OPEN	86	2	26.8	58.1	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D030	8	Main Road	9.520501	77.630678	29.5	118.4	MEDIUM	2	0.84	0.72	OPEN	49	5	31.2	18.9	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D031	20	Subramaniya Street	9.511698	77.61801	36.5	262.5	MEDIUM	2	0.92	0.88	OPEN	28	4	21.6	77.1	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D032	11	Nehru Street	9.50012	77.630886	23.8	126.4	HIGH	3	1.0	1.0	OPEN	26	1	5.9	24.6	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D033	31	Ayyappan Street	9.519243	77.619821	2.6	237.7	NONE	0	0.96	0.9	OPEN	84	5	5.2	65.1	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D034	7	Kamarajar Street	9.503471	77.638103	32.0	95.7	MEDIUM	2	1.13	0.67	OPEN	60	5	23.6	49.7	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D035	21	Tiruvalluvar Street	9.513316	77.635982	0.0	175.5	LOW	1	1.12	1.05	OPEN	34	1	34.4	27.2	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.

Drain_ID	Ward_No	Street_Name	Latitude	Longitude	Rainfall_mm_7day	Rainfall_mm_30day	Plastic_Accumulation_Level	Plastic_Accumulation_Score	Drain_Width_m	Drain_Depth_m	Drain_Type	Days_Since_Cleaning	Previous_Overflow_Count_1yr	Water_Level_cm	Blockage_Percent	Overflow_Risk_Label	Data_Source_Type	Notes
SVP-SYN-D036	32	Anna Street	9.503981	77.628997	39.2	123.0	LOW	1	1.11	0.65	OPEN	50	5	10.3	22.1	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D037	27	Perumal Street	9.506272	77.629648	106.2	237.7	LOW	1	1.02	1.17	OPEN	74	3	21.7	80.4	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D038	20	Subramaniya Street	9.519336	77.641836	0.0	244.7	LOW	1	0.88	0.73	OPEN	56	3	6.0	31.9	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D039	26	Sivan Street	9.502126	77.629685	79.0	260.2	NONE	0	0.72	0.94	OPEN	83	3	9.7	41.9	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D040	4	Sattur Road	9.507478	77.623292	17.5	299.9	LOW	1	1.0	1.1	OPEN	11	3	5.7	48.2	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D041	23	Patel Nagar	9.514254	77.637531	11.1	70.7	HIGH	3	0.56	0.61	OPEN	17	3	28.4	50.1	MEDIUM	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D042	14	Sastha Nagar	9.515487	77.637172	68.6	275.1	MEDIUM	2	0.49	1.09	COVERED	14	4	5.9	30.0	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D043	25	Rajaji Street	9.521537	77.622958	0.0	97.6	NONE	0	0.96	0.78	OPEN	16	4	32.4	32.8	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D044	28	Subramaniya Street	9.507264	77.630649	8.1	310.8	HIGH	3	0.48	1.18	OPEN	59	5	10.4	19.1	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.

Drain_ID	Ward_No	Street_Name	Latitude	Longitude	Rainfall_mm_7day	Rainfall_mm_30day	Plastic_Accumulation_Level	Plastic_Accumulation_Score	Drain_Width_m	Drain_Depth_m	Drain_Type	Days_Since_Cleaning	Previous_Overflow_Count_1yr	Water_Level_cm	Blockage_Percent	Overflow_Risk_Label	Data_Source_Type	Notes
SVP-SYN-D045	18	Chakrapani Street	9.518205	77.631454	7.8	188.9	MEDIUM	2	0.61	1.11	PARTIALLY_COVERED	12	2	35.9	24.5	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D046	30	Murugan Street	9.513258	77.627475	112.2	208.0	NONE	0	0.57	1.08	OPEN	77	5	35.8	49.5	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D047	13	Mariamman Kovil Street	9.504034	77.628161	20.6	159.8	LOW	1	0.97	0.87	OPEN	38	1	19.2	24.5	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D048	24	Nehru Street	9.511837	77.626594	89.2	177.9	LOW	1	0.76	0.78	OPEN	25	2	9.2	47.9	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D049	12	Amman Kovil Street	9.50341	77.630105	29.7	0.0	LOW	1	0.63	0.66	OPEN	30	2	11.3	6.1	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.
SVP-SYN-D050	9	Netaji Road	9.499138	77.619363	23.0	365.4	MEDIUM	2	1.0	0.66	OPEN	37	3	21.8	32.3	LOW	SYNTHETIC	Synthetic prototype data for ML demonstration only. Not actual field observations.

Sheet: README_Dataset_Documentation

Workbook: Srivilliputhur_Complete_Dataset_Layer1_Layer2(1).xlsx | Rows: 7 | Columns: 2

Srivilliputhur Plastic Waste & Drainage Overflow Prediction System — Complete Dataset	Column_2
DATASET STRUCTURE	This dataset is organized in two layers to maintain clear separation between real/public evidence and synthetic prototype data.
Layer 1: Real/Public Evidence	Sheets 1A–1F contain historical incidents, verified drainage records, municipality statistics, ward data, rainfall portals, and plastic/solid-waste information from publicly available sources. All records are labeled HISTORICAL_PUBLIC or VERIFIED_PUBLIC.
Layer 2: Synthetic Prototype ML Dataset	Sheet 2 contains 50 synthetic drain observations created for ML prototype demonstration only. All records are labeled SYNTHETIC. These do NOT represent actual field observations from Srivilliputhur.
IMPORTANT DISCLAIMER	Do NOT mix Layer 1 historical incident rows with Layer 2 synthetic drain rows in the same ML training set. They have different data-generating processes and units. Train ML models only on Layer 2 synthetic data for prototype demonstration.
How to cite Layer 1	Historical incidents: Cite news sources (Dinamalar, Samayam Tamil, OneIndia Tamil, Hindu Tamil). Municipality statistics: Cite SFD Lite Report, TN Urban Local Bodies. Ward data: Cite Census 2011.
How to cite Layer 2	Synthetic data: Clearly state in reports: The proposed ML model was developed and evaluated using synthetic drain-level data because publicly available Srivilliputhur records do not contain sufficient drain-level observations for supervised learning. Public historical and infrastructure data were used to establish the real-world context and inform the prototype feature design.
Future work	Replace Layer 2 synthetic records with actual field survey data collected from Srivilliputhur drains. Retrain model on real observations. Validate predictions against actual overflow events.

Sheet: Drain_Survey

Survey_Point_ID	Drain_ID	Survey_Date	Survey_Time_JST	Surveyor_ID	Ward_No	Street_or_Location	Landmark	Latitude_WGS84	Longitude_WGS84	GPS_Accuracy_m	GPS_Method	Drain_Present	Drain_Type	Drain_Cover_Status	Drain_Shape	Drain_Length_Observed_m	Drain_Width_m	Drain_Depth_m	Water_Depth_cm	Flow_Observed	Plastic_Level	Plastic_Description	Other_Desire_Level	Sediment_Depth_cm	Blockage_Percent	Blockage_Method	Overflow_Observed	Overflow_Evidence	Overflow_Severity	Last_Cleaning_Date	Cleaning_Info_Source	Days_Since_Cleaning	Photo_Overview_ID	Photo_CrossSection_ID	Photo_Blockage_ID	Addition_Photo_IDS	Weather_At_Visit	Rainfall_Recent_Notes	Notes	Record_Status	Data_Quality_Check	Column_4_3	_2	_3	_4	_5	_6										
SVP-SUR-001						Bustand area					Phone GPS																																	PLANNED													
SVP-SUR-002						Uzhavar Sant hai (Farmers Market) area					Phone GPS																																				PLANNED										
SVP-SUR-003						Periya Mari aman Kovi l area					Phone GPS																																				PLANNED										
SVP-SUR-004						Keezha Rath a Vedhi (Lower Car Street)					Phone GPS																																			PLANNED											
SVP-SUR-005						Vad akku Rath a Vedhi (North Car Street) junction					Phone GPS																																							PLANNED							

Survey_Point_ID	Drain_ID	Survey_Date	Survey_Time_IST	Surveyor_ID	Ward_No	Street_or_Location	Landmark	Latitude_WGS84	Longitude_WGS84	GPS_Accuracy_m	GPS_Method	Drain_Present	Drain_Type	Drain_Cover_Status	Drain_Shape	Drain_Length_Observed_m	Drain_Width_m	Drain_Depth_m	Water_Depth_cm	Flow_Observed	Plastic_Level	Plastic_Description	Other_Details_Level	Sediment_Depth_cm	Blockage_Percent	Blockage_Method	Overflow_Observed	Overflow_Evidence	Overflow_Severity	Last_Cleaning_Date	Cleaning_Info_Source	Days_Since_Cleaning	Photo_Overview_ID	Photo_CrossSection_ID	Photo_Blockage_ID	Additional_Photos	Weather_At_Visit	Rainfall_Recent_Notes	Notes	Record_Status	Data_Quality_Check	Column_43	_2	_3	_4	_5	_6					
SVP-SUR-006						Madavarvilagam Vaitthyanatha Swamy Temple area					Phone GPS																																PLANNED									
SVP-SUR-007						Raitanpatti area					Phone GPS																																		PLANNED							
SVP-SUR-008						Kullar Street					Phone GPS																																		PLANNED							
SVP-SUR-009						Ottamadam					Phone GPS																																		PLANNED							
SVP-SUR-010						Kottaiapatti					Phone GPS																																		PLANNED							
SVP-SUR-011						Ashok Nagar					Phone GPS																																			PLANNED						
SVP-SUR-012						GM Nagar					Phone GPS																																			PLANNED						
SVP-SUR-013						Sandhai Kinaru Street					Phone GPS																																			PLANNED						
SVP-SUR-014						Mudaliarpatti Street					Phone GPS																																				PLANNED					
SVP-SUR-015						Chettiyakudi Street					Phone GPS																																			PLANNED						
SVP-SUR-016						Pettalkadai Street					Phone GPS																																				PLANNED					

[illegible]

Survey_Point_ID	Drain_ID	Survey_Date	Survey_Time_IST	Surveyor_ID	Ward_No	Street_or_Location	Landmark	Latitude_WGS84	Longitude_WGS84	GPS_Accuracy_m	GPS_Method	Drain_Present	Drain_Type	Drain_Cover_Status	Drain_Shape	Drain_Length_Observed_m	Drain_Width_m	Drain_Depth_m	Water_Depth_cm	Flow_Observed	Plastic_Level	Plastic_Description	Other_Debris_Level	Sediment_Depth_cm	Blockage_Percent	Blockage_Method	Overflow_Observed	Overflow_Evidence	Overflow_Severity	Last_Cleaning_Date	Cleaning_Info_Source	Days_Since_Cleaning	Photo_Overview_ID	Photo_CrossSection_ID	Photo_Blockage_ID	Additional_Photos	Weather_At_Visit	Rainfall_Recent_Notes	Notes	Record_Status	Data_Quality_Check	Column_43	_2	_3	_4	_5	_6			
SVP-SUR-023						Comparison point: select an accessible drain after field route check					Phone GPS																																PLANNE D							
SVP-SUR-024						Comparison point: select an accessible drain after field route check					Phone GPS																																	PLANNE D						
SVP-SUR-025						Comparison point: select an accessible drain after field route check					Phone GPS																																		PLANNE D					

Sheet: README

Workbook: Srivilliputhur_Drainage_Field_Survey_Template (1)(1).xlsx | Rows: 12 | Columns: 2

Srivilliputhur Drainage Field Survey Template	Column_2
Purpose	Collect primary, real-world observations for the AI-Based Plastic Waste & Drainage Overflow Prediction System. Do not enter estimated coordinates from online sources.
How to use	Each row represents one actual drain segment or survey point. A planned point becomes a completed point only after GPS, observation and photo evidence are recorded.
Survey_Point_ID	Permanent visit/planning identifier. Keep the prefilled ID. For a new site use SVP-SUR-026, SVP-SUR-027, etc.
Drain_ID	Assign only after a drain is physically verified. Suggested permanent format: SVP-Wxx-D001. If Ward_No is not verified, use SVP-WUN-D001 and update only after verification, retaining an ID-change log.
GPS	Stand near the observed drain segment, enable phone location, record latitude/longitude in decimal degrees (WGS84), GPS accuracy in metres, and use Phone GPS as method. Never copy approximate web coordinates.
Measurement	Measure visible drain length, width and depth in metres. Record water depth and sediment depth in centimetres. Write NOT_AVAILABLE when inaccessible.
Plastic and debris	Use NONE/LOW/MEDIUM/HIGH based on the rubric sheet. Add a short description: e.g., bottles, bags, food wrappers, leaves, construction waste. Do not enter kg unless a separate approved weighing protocol is used.
Blockage percent	Estimate 0–100 based on the visible cross-section blocked by debris/silt/waste. Record Blockage_Method. If you cannot observe safely, write NOT_AVAILABLE.
Photos	Use a consistent filename. Example: SVP-SUR-001_2026-09-16_overview_01.jpg. Store photos in folders by date and do not photograph people without consent.
Safety	Do not enter drains, handle sharp waste, work in heavy rain/overflow, obstruct traffic, or survey alone. Obtain permission where required.
Data status	Use PLANNED before visit; COMPLETE only after checking all required fields; REVISIT_REQUIRED when a key measurement/GPS/photo is missing.
Minimum complete record	Date, time, surveyor, location, latitude, longitude, drain present, drain type, width/depth if measurable, plastic level, blockage %, overflow, at least one photo ID, and notes.

Sheet: Scoring_Rubric

Workbook: Srivilliputhur_Drainage_Field_Survey_Template (1)(1).xlsx | Rows: 6 | Columns: 2

Field	Allowed values / recommended rubric
Plastic_Level	NONE = no visible plastic; LOW = isolated items, less than about 25% of visible channel length/area; MEDIUM = noticeable accumulation, about 25–50%; HIGH = extensive accumulation, more than 50%. This is an observation class, not mass.
Other_Debris_Level	Use same NONE/LOW/MEDIUM/HIGH scale for leaves, silt, construction debris and mixed waste excluding plastic.
Blockage_Percent	0 = clear; 1–24 = low obstruction; 25–49 = moderate; 50–74 = substantial; 75–100 = severe/nearly blocked. Estimate visible hydraulic cross-section, not just surface litter.
Overflow_Severity	NONE = no evidence; MINOR = local puddling/edge flow; MODERATE = roadway/compound affected; SEVERE = widespread inundation or access affected. Record source/evidence in notes.
Drain_Type	OPEN = uncovered channel; COVERED = fully covered; PARTIALLY_COVERED = intermittent cover; CLOSED = enclosed/non-visible system.
Data_Quality_Check	COMPLETE = all minimum fields valid; GPS_RECHECK = coordinate/accuracy problem; PHOTO_MISSING = evidence missing; MEASUREMENT_UNAVAILABLE = access/visibility issue; REVIEW_REQUIRED = other issue.

Final Consolidated Data Status

The consolidated materials contain real/public source evidence, historical/public records, a synthetic 50-row ML prototype dataset, and a 25-point field-survey template. The field-survey rows are templates/planned observations unless completed with actual GPS, observations and evidence. The complete municipal drain inventory, complete drain-level condition/maintenance observations, and a verified Srivilliputhur station daily rainfall series remain unavailable in the supplied materials.

Key rule: synthetic ML data must remain explicitly labeled synthetic prototype data.